QBUS2820 Assignment 2 (25 marks)

October 16, 2026

1 Background and Task

The Consumer Price Index (CPI) tracks the average price changes of a basket of consumer goods and services, including transportation, food, and medical care. By calculating the weighted average of these price changes, the CPI provides a reliable measure of inflation or deflation, reflecting shifts in the cost of living over time. As one of the most commonly used economic indicators, CPI offers valuable insights into broader price trends and economic conditions.

In this project, your goal is to develop a predictive model to forecast CPI of a particular sector given its historical quarterly values. The CPI dataset `CPI_train.csv` contains the quarterly CPI data from March 2000 to December 2024 (100 data points). This dataset is based on a real CPI dataset with some added noise for de-identification purposes. The test dataset `CPI_test.csv` (not provided) has the same structure as the training data and contains the quarterly CPI data from March 2025 to December 2026 (8 data points).

To be specific, your task is to develop a time series model using `CPI_train.csv` to forecast the quarterly CPI measures from March 2025 to December 2026. Note that this is a **multiple-step-ahead forecast problem**.

1.1 Test Error

Prediction accuracy will be measured using mean squared error (MSE) on the test data. Let $\hat{y}_{T+h|1:T}$ be the h-step-ahead forecast of y_{T+h}, based on the training data $y_{1:T}$, where y_{T+h} is the h-th value in the test data `CPI_test.csv`. The test error is

$$\text{Test error} = \frac{1}{8}\sum_{h=1}^{8}\left(\hat{y}_{T+h|1:T} - y_{T+h}\right)^2,$$

where 8 is the number of observations in the test dataset.

2 Submission Instructions

Please submit the following **TWO** files through Canvas. You may submit additional files if they are required by your implementation.

1. A Jupyter Notebook named **SID_implementation.ipynb**, where "SID" is replaced by your student ID, that implements your data analysis and produces the test error when run by the marker.

2. A CSV file named `CPI_forecast.csv` containing the 8 quarterly CPI forecasts made by your final predictive model, from March 2025 to December 2026. The file must contain two columns: the first column named `Quarter`, indicating the quarter, and the second column named `CPI`, containing the predicted CPI for that quarter.

If additional files are required, use the format **SID_<name>**, with your student ID and a short descriptive filename.

2.1 Notebook Requirements

Assume that `CPI_train.csv` is in the same folder as your notebook. The marker will place the test file `CPI_test.csv` in that folder before running the notebook. If model training involves generating random numbers, fix the random seed in **SID_implementation.ipynb**, for example using `np.random.seed(0)`, so that the marker can reproduce your results.

The following code must appear in the **last code cell** of your submitted notebook:

```
import pandas as pd

CPI_test = pd.read_csv("CPI_test.csv")

# YOUR CODE HERE: code that produces the test MSE test_error

print(test_error)
```

The final cell must calculate the same test MSE that you would obtain if you were given the test data. Your notebook must include clear explanations and comments so that the marker can understand and run your approach.

You may use any methods and publicly available Python libraries to implement your models.

Your notebook must describe your data analysis procedure in enough detail for another data scientist with relevant background knowledge to understand and reproduce your analysis. It should be clearly structured, with organised sections and subsections, use Markdown cells to explain the code, and present relevant visualisations clearly.

2.2 Acknowledging the Use of AI

Generative AI tools may be used to support your learning and work, for example to clarify concepts, help debug code, or improve the clarity of your writing. However, the submitted notebook must demonstrate your own understanding, analytical decisions, and interpretation of the results. **You must not submit a Jupyter Notebook generated entirely by AI.** Making superficial edits to an AI-generated notebook does not meet this requirement.

You are responsible for checking the correctness and appropriateness of any AI-assisted content. You must be able to explain the code, justify the modelling decisions, and interpret the results in your submission. Acknowledging AI use does not replace these responsibilities.

You must include a statement at the end of your submission indicating whether or not generative AI tools were used.

- If you did not use generative AI tools, state clearly that no AI assistance was used.
- If you used generative AI tools, state the name and version of each tool, its publisher, and how it was used in completing the assignment.

Up to 2 marks will be deducted if the required statement is missing.

You are also required to maintain a record of your AI use during the assignment and retain copies of all AI-generated outputs as evidence of your research and writing process.

3 Marking Criteria

This assignment is worth 25 marks. Fourteen marks are allocated to the presentation of the Jupyter Notebook, including the description of your data analysis procedure, and 11 marks to prediction accuracy.

1. **Prediction accuracy and implementation (11 marks).**

   The marker will run **SID_implementation.ipynb**. The test MSE will be compared with the smallest test MSE among all submissions, including models developed by the teaching team.

   - If the notebook runs successfully and produces a test MSE, up to 11 marks will be awarded based on prediction accuracy and the appropriateness of the implementation.
   - If the notebook cannot be run successfully or does not produce a test MSE, partial marks of at most 3 may be awarded based on the appropriateness of the submitted implementation.

2. **Notebook presentation and data analysis procedure (14 marks).**

   Marks will be awarded for the readability and organisation of the notebook, the appropriateness of the forecasting method, and the quality, detail, discussion, and explanation of the data analysis.

3. **Prediction CSV.**

   Up to 2 marks will be deducted if the prediction CSV is missing or does not follow the required format.

4. **Late submission.**

   The late penalty is 5% of the assigned mark per calendar day. The closing date is the last date on which the assignment will be accepted for marking. Assignments submitted after the closing date will be assigned a mark of zero.

4 Special Consideration

If you have been granted special consideration with an extended due date later than the original closing date, your new closing date will be the same as your extended due date. No submission will be accepted after that date. Failure to submit by the extended due date will result in a mark of zero.

5 Final Notes

Read all instructions carefully and seek clarification early if you are unsure about any requirement. If you believe there is an error in the assignment, contact the teaching team as soon as possible.